

Table 1. Effectiveness of the Approaches on Benchmark Libraries

Library			Dont-break			NoREGRETS+			TsINFER-NODE		
Name	Updates	Dependents	# TP	# FP	# FN	# TP	# FP	# FN	# TP	# FP	# FN
<i>async</i> 2.0.0	4	34,268	1	0	11	3	5	9	2	6	10
<i>execa</i> 3.0.0	6	13,791	0	0	13	6	4	7	6	4	7
<i>jsonwebtoken</i> 2.0.0	7	32,658	0	0	2	0	1	2	0	0	2
<i>find-up</i> 3.0.0	39	5,304	0	0	2	1	0	1	0	0	2
<i>camelcase</i> 1.0.0	26	7,305	0	0	0	0	2	0	0	1	0
<i>moment</i> 2.0.0	5	72,419	0	0	0	1	0	0	0	1	1
<i>qs</i> 1.0.0	60	17,515	0	1	0	0	2	0	0	2	0
<i>path-to-regexp</i> 1.0.0	55	6,764	0	0	0	0	2	0	0	2	0
<i>has-flag</i> 1.0.0	5	3,281	0	0	0	0	2	0	0	2	0
<i>uuid</i> 2.0.0	16	68,361	1	0	0	1	0	0	1	0	0
<i>days</i> 2.0.0	126	20,310	0	0	2	0	1	2	1	3	1
<i>fast-deep-equal</i> 1.0.0	14	5,064	0	0	2	1	0	1	1	1	1
<i>strip-bom</i> 1.0.0	10	2,107	0	0	2	0	3	2	1	3	1
<i>joi</i> 9.0.0	56	12,851	2	0	38	2	4	38	3	7	37
<i>immutable</i> 1.0.0	46	8,653	2	0	15	2	6	15	8	3	9
<i>bignumber.js</i> 1.0.0	54	8,432	0	0	0	2	0	0	0	0	0
<i>extend</i> 1.0.0	161	5,379	0	0	0	0	0	0	0	0	0
<i>is-plain-object</i> 1.0.0	13	2,585	0	0	0	0	0	0	0	0	0
<i>underscore</i> 1.0.0	54	23,640	0	0	0	0	1	0	0	2	0
<i>fill-range</i> 1.0.0	33	2,476	0	0	1	0	1	0	0	0	0
<i>mongoose</i> 3.0.0	43	19,597	0	0	44	3	4	41	2	3	42
<i>p-limit</i> 1.0.0	18	3,878	0	0	0	0	0	0	0	0	0
<i>decimal.js</i> 1.0.0	51	3,094	0	0	0	2	0	0	0	2	0
<i>minimist</i> 1.0.0	14	24,763	0	0	0	0	2	0	0	0	0
<i>throttle-debounce</i> 1.0.0	15	3,472	0	0	0	0	0	0	0	0	0
<i>jsonpath</i> 2.0.0	4	1,475	0	0	0	0	0	0	0	0	0
<i>pluralize</i> 1.0.0	21	5,574	0	0	0	0	0	0	0	0	0
<i>less</i> 1.0.0	49	9,779	0	0	0	0	3	0	0	3	0
<i>express</i> 3.0.0	27	85,792	0	0	16	2	0	14	0	0	16
<i>clean-stack</i> 1.0.0	14	24,406	0	0	0	0	2	0	0	0	0
<i>flatted</i> 1.0.0	31	2,390	0	0	0	0	0	0	0	0	0
<i>js-tokens</i> 1.0.0	4	5,798	0	0	4	0	0	4	0	0	4
<i>urlpattern-polyfill</i> 1.0.0	22	134	0	0	0	0	0	0	0	1	0
<i>dot-drop</i> 1.0.0	23	1,647	0	0	0	0	0	0	0	0	0
<i>winston</i> 2.0.0	24	23,835	0	0	19	2	1	17	2	1	17
<i>spdx-correct</i> 1.0.0	13	1,055	0	0	0	0	1	0	0	0	0
<i>ioredis</i> 1.0.0	21	6,671	0	0	0	0	0	0	0	0	0
<i>memfs</i> 2.0.0	136	696	0	0	0	0	3	0	0	0	0
<i>filesize</i> 1.0.0	124	2,314	0	0	0	0	0	0	0	0	0
<i>psl</i> 1.0.0	53	1,023	0	0	0	0	0	0	0	0	0
<i>pretty-format</i> 2.0.0	3	1,600	0	0	0	0	0	0	0	0	0
Total			9	1	167	19	41	157	16	46	160